

KARAN RAJ SINGH

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PROFESSIONAL SUMMARY

Computer Science graduate with hands-on experience building full-stack web applications using React.js on the frontend and Python (FastAPI) REST APIs on the backend, including JWT authentication and both relational (PostgreSQL) and NoSQL (MongoDB) database integration. Experienced working in Agile, cross-functional teams from design through deployment. Comfortable across the stack — from responsive, mobile-first React interfaces to manual/API testing with Postman and version control with Git/GitHub — with additional hands-on exposure to LLM integration and RAG pipelines.

SKILLS

Languages: JavaScript, Java, Python

Frontend: HTML, CSS, React.js

Backend: Node.js, Express.js, FastAPI, Pydantic, RESTful API, JWT Authentication

AI & GenAI: LLM APIs, RAG, Vector Embeddings, Vector Database

Databases: MongoDB, PostgreSQL, MySQL, pgvector, Redis

Tools: Git, GitHub, pytest, Postman, VS Code, IntelliJ IDEA

Projects

Personal Task Manager | *React.js, Node.js, Express.js, MongoDB, JWT, LLM API*

- Built a full-stack task manager with 8 RESTful endpoints covering CRUD, search, filtering, sorting, and pagination, backed by MongoDB with per-user data isolation enforced at the middleware layer.
- Implemented JWT-based auth with bcrypt password hashing and rate-limited login endpoints to prevent brute-force attempts, protecting all authenticated routes end-to-end.
- Integrated a third-party LLM API (Gemini function-calling) to convert natural-language input (“remind me to call the dentist Friday”) into structured task objects, cutting manual task entry to a single sentence.
- Developed a responsive React frontend with Redux Toolkit for centralized state across Dashboard, Tasks, and Settings modules, with optimistic UI updates for instant task actions.

Resume Analyzer | *React, React Router, Python, FastAPI, GenAI*

- Built a full-stack AI resume analyzer and interview coach with a React.js frontend and Python (FastAPI) RESTful backend, parsing resumes and job descriptions (PDF/DOCX via PyMuPDF) into structured data and producing a weighted ATS compatibility score with section-by-section justification.
- Modeled and queried a relational PostgreSQL database (with pgvector) for semantic resume–JD similarity search not keyword overlap generating an overall match score plus skills/experience/projects/education sub-scores and a ranked skill-gap breakdown.
- Designed a RAG-based career assistant and AI resume rewriter using direct LLM API calls (no framework abstractions): a hand-built retrieval-then-generate pipeline answers grounded questions about the user's own resume/JD data, while a role-adaptive rewriter turns weak bullet points into metric-driven ones.

Education

UTTARANCHAL UNIVERSITY, DEHRADUN
B.TECH, Computer Science and Engineering

2022-2026
CGPA: 6.65